

TABLE V
RESAMPLER PARAMETERS AND VALUES

Parameter	Value
Interpolation	48
Decimation	50
Taps	-
Fractional bandwidth	0

TABLE VI
MULTIPLY CONSTANT BLOCK

Parameter	Value
Constant	Controlled by slider

TABLE VII
AUDIO SINK BLOCK

Parameter	Value
Constant	48kHz

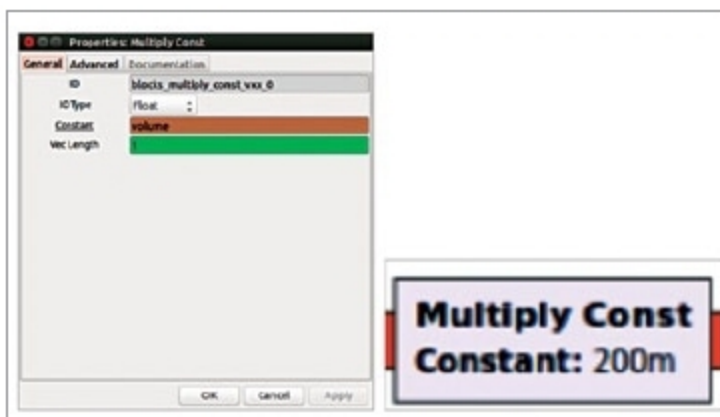


Fig. 11: Multiply constant block for volume settings

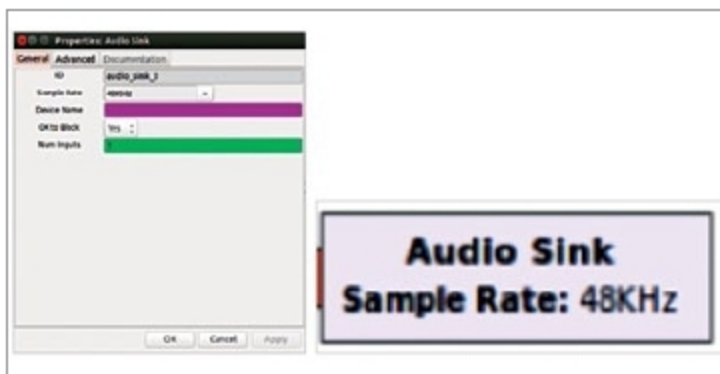


Fig. 12: Audio sink block value and settings

from the tool library in GNU Radio Companion (GRC). It is available on the right side of the window. Then, change the parameters of the block by double clicking it as shown in Table I.

Rational resampler block. This block when added to the design provides signal changes and makes it suitable to provide as input to further blocks. Rational resampler block

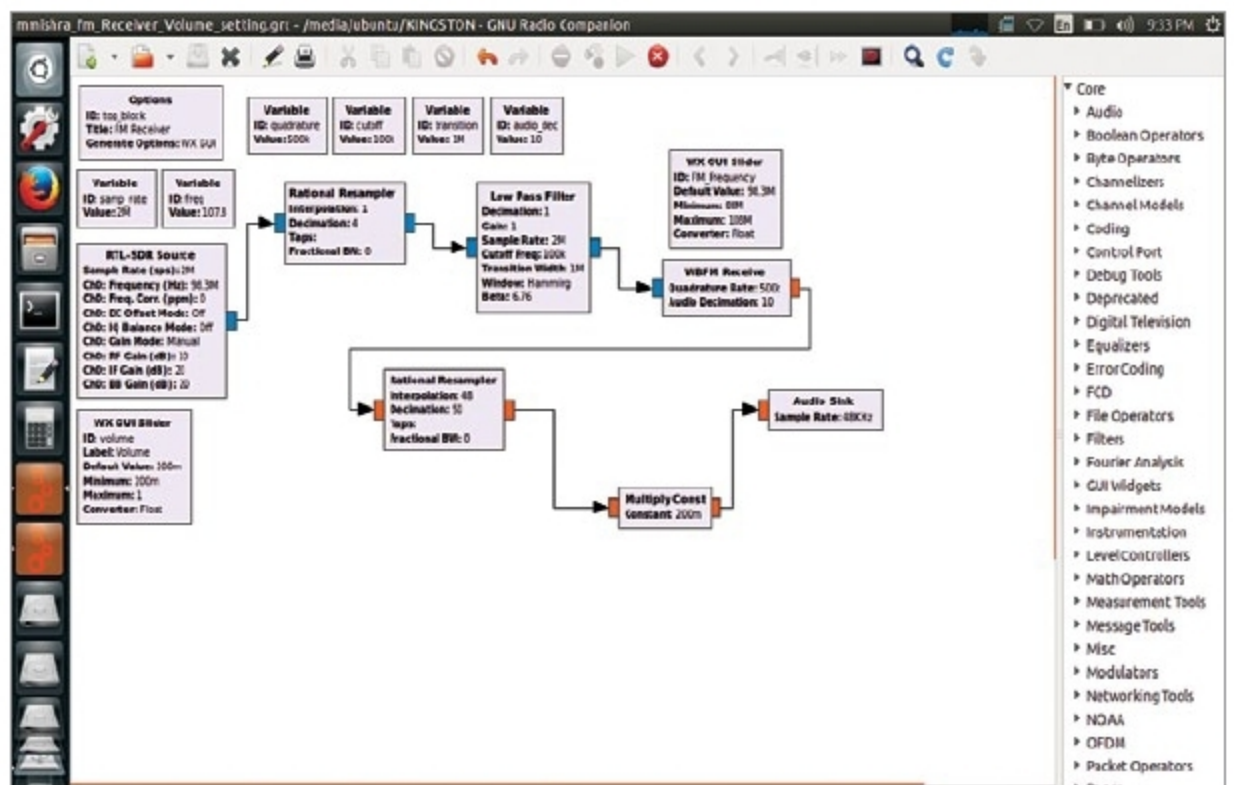


Fig. 13: Radio receiver flowgraph designed in GNU Radio

settings are shown in Fig. 7, and its parameters are given in Table II.

Low-pass filter block. Low-pass filter block is added after setting rational resampler block. Values of various parameters for a low-pass filter are given in Table III.

WBFM receive block. This block is placed to extract demodulated signal from incoming signal. Parameters and values are given in Table IV.

Rational resampler after WBFM receive block. This block provides functionality of changing the signal to make it suitable for audio output block. Values are given in Table V.

Multiply constant block. This block is used for controlling the audio volume using GUI of FM receiver implemented using GNU Radio. Details of multiply constant block are given in Table VI.

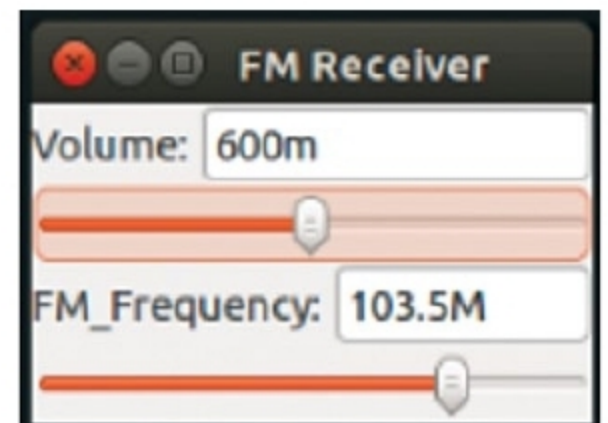


Fig. 14: FM receiver designed in GNU Radio interface

Audio sink block. This block provides audio output signal at the computer speakers. Details of audio sink block are shown in Fig. 12 and also listed in Table VII.

Run implemented receiver using GNU Radio. The receiver is designed using two sliders for controlling frequency and volume. Complete GRC code (FM_Receiver_gnu.grc) is shown in Fig. 13, and FM receiver window is shown in Fig. 14. Frequency range supported is 88MHz to 108MHz. The slider can set the frequency between 88MHz and 108MHz, with 0.1MHz step size. When the slider is used to control a value in runtime, give variable name as the slider identifying value in parameter of the particular block. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com



Madhuram Mishra is pursuing ME (digital communication) from NITTTR, Bhopal. He has keen interest in circuit design and Arduino boards